Dual-Tank Autonomous Water Level Monitoring and Motor Control System

Task 3 – Automation System Design | Tool: Tinkercad (Arduino UNO) | CodeAlpha Robotics & Automation Internship

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1. Introduction

The **Dual-Tank Autonomous Water Level Monitoring and Motor Control System** is a fully automated embedded system that monitors water levels in two tanks simultaneously and controls a pump motor without any human intervention. Designed and simulated using **Tinkercad** with an Arduino UNO, the system uses HC-SR04 ultrasonic sensors to measure water levels in both an upper storage tank and a lower source tank. The motor is turned ON or OFF automatically based on predefined thresholds, and real-time status is shown on a 16x2 LCD screen.

Key advantage: Unlike basic single-tank controllers, this system checks the lower (source) tank before activating the motor — preventing pump dry-run damage while ensuring the upper tank is always filled efficiently.

2. Components & Pin Mapping

Component	Specification	Purpose
Arduino UNO	ATmega328P, 5V	Main microcontroller
HC-SR04 (x2)	Range: 2–400 cm	Measure water level (upper & lower tank)
TIP120 Transistor	NPN Darlington	Motor switching via base resistor
DC Motor	5V–9V	Simulates water pump
16x2 LCD	HD44780, 5V	Real-time status display
9V Battery	DC Supply	External power for motor
Resistor 1kOhm	1000 Ohm	Base resistor for TIP120
Potentiometer	10 kOhm	LCD contrast adjustment

Pin Connections: Upper HC-SR04 → Pins 10/11 | Lower HC-SR04 → Pins 8/9 | Motor (TIP120 base) → Pin 6 | LCD → Pins 2,3,4,5,12,13 | Potentiometer → LCD contrast (V0 pin)

3. Working Principle & System Flowchart

Each HC-SR04 sends a 10 µs ultrasonic pulse and the Arduino measures the return echo. Distance is calculated as: Distance (cm) = Echo Duration (µs) x 0.034 / 2. The system checks conditions in priority order every 1 second:

#	Condition	Motor	LCD Output
1	Lower tank empty (dist > 9 cm)	**OFF**	LOWER TANK EMPTY
2	Upper tank full (dist < 5 cm)	**OFF**	TANK FULL MotorOFF
3	Upper tank empty (dist > 18 cm)	**ON**	Motor: ON (Filling)

#	Condition	Motor	LCD Output
4	Intermediate level	**OFF**	Tank OK Level

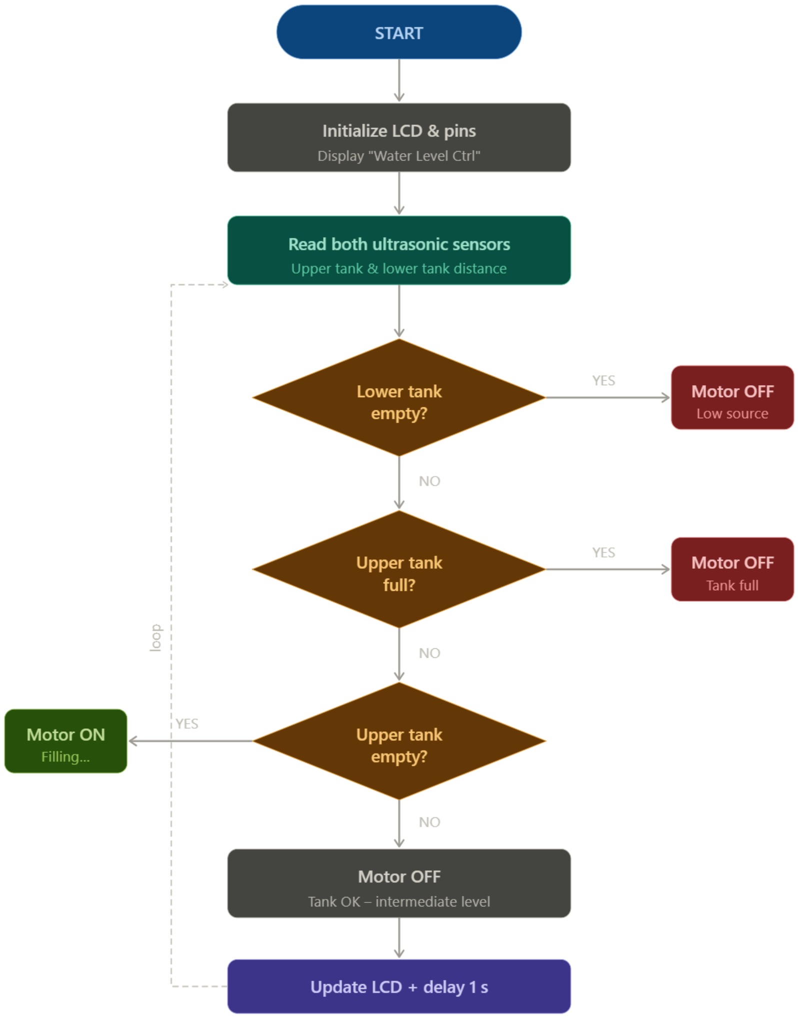

Figure 1: System Decision Flowchart

4. Circuit Description & Circuit Diagram

- **Sensors:** Two HC-SR04 modules positioned at the top of each tank. Upper tank sensor on pins 10/11; lower on 8/9. Distance measured downward to water surface.
- **Motor Driver:** TIP120 NPN Darlington transistor controlled by Pin 6 through a 1kOhm base resistor. The 9V battery powers the motor; Arduino controls only the gate signal.
- **LCD Display:** 16x2 display in 4-bit mode on pins 2,3,4,5,12,13. Potentiometer adjusts contrast on V0 pin. Shows real-time sensor distances and motor status.
- **Tinkercad Project Link:** https://www.tinkercad.com/things/hx9BEGO5ihI-smart-dual-tank-water-level-controller

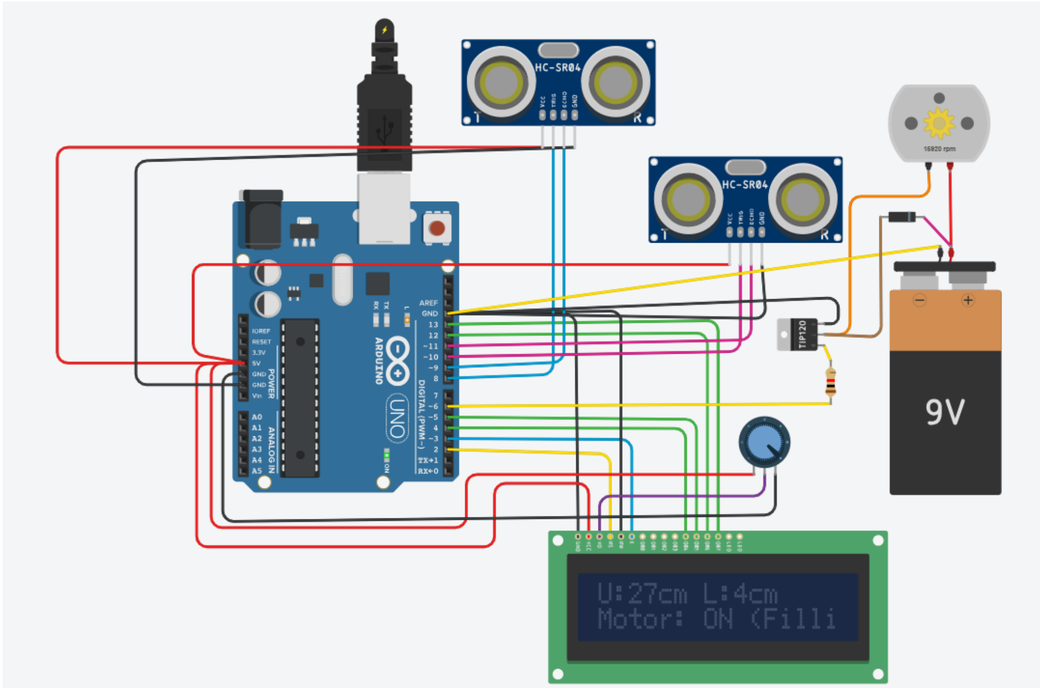

Figure 2: Circuit Diagram

5. Applications & Conclusion

Applications: Residential rooftop tank filling, agricultural irrigation control, industrial process water management, and smart building water systems.

Benefits: Prevents water overflow and wastage, protects pump from dry-run damage, reduces manual monitoring effort, and can be extended with IoT (ESP8266/ESP32) for remote alerts.

This project successfully demonstrates a fully autonomous dual-tank water level control system. All four operating states were validated in Tinkercad simulation. The system is practical, cost-effective, and ready for real-world deployment with minimal hardware modifications.

References

[1] Arduino Documentation: https://docs.arduino.cc/hardware/uno-rev3

[2] HC-SR04 Datasheet: https://cdn.sparkfun.com/datasheets/Sensors/Proximity/HCSR04.pdf

[3] TIP120 Datasheet: https://www.onsemi.com/pdf/datasheet/tip120-d.pdf

[4] Autodesk Tinkercad Simulator: https://www.tinkercad.com